b230e4.c

b230e4.c > main()

```c
#include <stdio.h>

#define MAX 5

int deque[MAX];
int left = -1, right = -1;

void insert_left(int val);
void insert_right(int val);
void delete_left(void);
void delete_right(void);
void display(void);

void insert_right(int val)
{
    if((left == 0 && right == MAX - 1) || (left == right + 1))
    {
        printf("\nOVERFLOW: Cannot insert %d\n", val);
        return;
    }
    if (left == -1)
    {
        left = 0;
        right = 0;
    }
    else
    {
        if(right == MAX - 1)
            right = 0;
        else
            right = right + 1;
    }
    deque[right] = val;
    printf("Inserted Normal Priority Sensor Reading: %d%%\n", val);
}
```

```c
void insert_right(int val)

void insert_left(int val)
{
    if((left == 0 && right == MAX - 1) || (left == right + 1))
    {
        printf("\nOVERFLOW: Cannot insert %d\n", val);
        return;
    }

    if (left == -1)
    {
        left = 0;
        right = 0;
    }
    else
    {
        if(left == 0)
            left = MAX - 1;
        else
            left = left - 1;
    }
    deque[left] = val;
    printf("Inserted HIGH Priority Sensor Reading: %d%%\n", val);
}

void delete_left()
{
    if (left == -1)
    {
        printf("\nUNDERFLOW: Queue is Empty\n");
        return;
    }
    printf("Processed Sensor Reading from Left: %d%%\n", deque[left]);
    if(left == right)
    {
        left = -1;
```

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```
61  void delete_left()
69      if(left == right)
72          right = -1;
73      }
74      else
75      {
76          if(left == MAX - 1)
77              left = 0;
78          else
79              left = left + 1;
80      }
81  }
82
83  void delete_right()
84  {
85      if (left == -1)
86      {
87          printf("\nUNDERFLOW: Queue is Empty\n");
88          return;
89      }
90      printf("Removed Sensor Reading from Right: %d%%\n", deque[right]);
91      if(left == right)
92      {
93          left = -1;
94          right = -1;
95      }
96      else
97      {
98          if(right == 0)
99              right = MAX - 1;
100         else
101             right = right - 1;
102     }
103 }
104
105 void display()
```

```c
void display()
{
    int front = left, rear = right;
    if(front == -1)
    {
        printf("Current Deque: [ Empty ]\n\n");
        return;
    }
    printf("Current Deque Elements: ");

    if(front <= rear)
    {
        while(front <= rear)
        {
            printf("%d ", deque[front]);
            front++;
        }
    }
    else
    {
        while(front <= MAX - 1)
        {
            printf("%d ", deque[front]);
            front++;
        }
        front = 0;
        while(front <= rear)
        {
            printf("%d ", deque[front]);
            front++;
        }
    }
    printf("\n\n");
}

int main()
```

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```
105   void display()
123       else
131           while(front <= rear)
136       }
137       printf("\n\n");
138   }
139
140   int main()
141   {
142       printf("=== SUGARCANE IRRIGATION SYSTEM (DOUBLE ENDED QUEUE) ===\n\n");
143
144       printf("1. Inserting normal sensor readings at Right...\n");
145       insert_right(35);
146       display();
147
148       insert_right(28);
149       display();
150
151       printf("2. Inserting urgent critical reading at Left (High Priority)...\n");
152       insert_left(12);
153       display();
154
155       printf("3. Processing sensor data from Left...\n");
156       delete_left();
157       display();
158
159       printf("4. Removing outdated reading from Right...\n");
160       delete_right();
161       display();
162
163       return 0;
164   }
```

```
=== SUGARCANE IRRIGATION SYSTEM (DOUBLE ENDED QUEUE) ===

1. Inserting normal sensor readings at Right...
Inserted Normal Priority Sensor Reading: 35%
Current Deque Elements: 35

Inserted Normal Priority Sensor Reading: 28%
Current Deque Elements: 35 28

2. Inserting urgent critical reading at Left (High Priority)...
Inserted HIGH Priority Sensor Reading: 12%
Current Deque Elements: 12 35 28

3. Processing sensor data from Left...
Processed Sensor Reading from Left: 12%
Current Deque Elements: 35 28

4. Removing outdated reading from Right...
Removed Sensor Reading from Right: 28%
Current Deque Elements: 35
```